



Human brainstem activation underpinning offset analgesia changes in perceived pain intensity. An ultra-high field functional magnetic resonance imaging investigation

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ABSTRACT

Offset analgesia is a pain modulatory phenomenon whereby a slight decrease in the intensity of a painful stimulus results in a large, disproportionate reduction in perceived pain. While human brain imaging studies have begun to map the brain circuitry underpinning offset analgesia, the brainstem circuits responsible for this phenomenon remain unknown. Furthermore, it is now recognised that previously described analgesic paradigms can evoke pain increases in some individuals, but whether the same brainstem circuitry is responsible for inhibitory and paradoxical facilitatory responses remains to be classified. In this exploratory brain imaging investigation, across 37 pain-free participants we used ultra-high field functional magnetic resonance imaging to define brainstem sites associated with offset pain intensity decreases and paradoxical increases. We found that in response to offset analgesia trials, 12 participants reported significant decreases; 8 reported significant increases; and 17 reported no significant change in perceived pain intensity during noxious heat stimulation of the forearm using a thermode. Brainstem-specific analysis showed offset pain decreases, associated with bilateral signal intensity decreases in the locus coeruleus, part of the rostral ventrolateral medulla, and the nucleus tractus solitarius. Signal increases were found in the parabrachial nucleus and the subcoeruleus region. In striking contrast, pain intensity increases were associated with signal decreases in the rostral ventromedial medulla and signal increases in the pontine reticular nucleus. No significant signal intensity changes were observed in other key brainstem pain modulation regions such as the periaqueductal gray matter and the A5 cell group. These data support the role of and identify specific brainstem nuclei function associated with offset pain intensity decreases and increases.

1. Introduction

While it is well-established that the human brain can profoundly modulate noxious inputs, the precise neural circuits underpinning this phenomenon remain to be defined. One experimental paradigm designed to explore analgesic ability in humans is offset analgesia, which is characterised by a disproportionate reduction in perceived pain following a slight decrease in the intensity of a painful stimulus (Fust

et al., 2021). The magnitude of offset analgesia has been shown to be significantly reduced in individuals with various forms of chronic pain (Oudejans et al., 2015; Shulman et al., 2020; Szikszay et al., 2019; Zhang et al., 2018; Larsen et al., 2022), an effect thought to result from altered functioning of endogenous central pain-modulation circuits. Given the high prevalence and limited effective treatment options for chronic pain sufferers (Disease et al., 2016; Backonja and Woolf, 2010), understanding the precise neural circuits that mediate pain modulation in

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humans could provide a better understanding of underlying pathophysiology, thereby helping develop treatments.

Using functional magnetic resonance imaging (fMRI), it has been reported that offset analgesia is associated with signal intensity changes in various brain regions including the anterior cingulate (ACC), dorso-lateral prefrontal cortex (dlPFC), medial prefrontal cortex (mPFC), insula, nucleus accumbens, and the dorsal and ventral pons (Zhang et al., 2018). These findings are consistent with a recent study using near infrared spectroscopy that reported offset analgesia was associated with altered activity in both the dlPFC and mPFC (Alter et al., 2022). Furthermore, multiple fMRI studies have shown that other pain modulatory paradigms such as placebo analgesia and conditioned pain modulation are associated with activity changes in forebrain regions including the ACC and dlPFC (Boorman et al., 2025; L.S. Crawford et al., 2023; L.S. Crawford et al., 2023; Youssef et al., 2016).

While it is increasingly clear that forebrain regions are involved in multiple pain modulatory phenomena, it is generally accepted that these regions modulate incoming noxious inputs thorough descending projections to brainstem circuits such as the midbrain periaqueductal gray (PAG) – rostral ventrolateral medulla (RVM) – dorsal horn/spinal trigeminal nucleus – pathway (Millan, 2002; Ossipov et al., 2010). Whilst an earlier fMRI study reported that offset analgesia was associated with signal changes in this PAG-RVM circuit, limited spatial resolution at the time meant that the precise locations of these changes were relatively imprecise (Derbyshire and Osborn, 2009). Given significant improvements in spatial resolution now available using both high field and ultra-high field fMRI, it is possible to redefine the location of the brainstem circuitry responsible for offset analgesia. Indeed, using this approach we showed recently, that placebo analgesia is associated with signal changes in the lateral PAG, rather than the ventrolateral PAG which had been presumed to be the PAG subregion mediating this analgesic effect (Crawford et al., 2021; L.S. Crawford et al., 2025).

This investigation aimed to leverage ultra-high field fMRI to identify the precise brainstem neural circuits underpinning *offset* pain intensity changes. We hypothesised: (i) that *offset* pain intensity changes would evoke signal changes in the PAG and RVM, and (ii) that brainstem signal changes would be most pronounced during the final portion of the *offset* stimulus, during the return to moderate intensity following a minor noxious stimulus intensity increase. Defining this pain modulatory circuitry is important for a better understanding of the role of these circuits in the development and maintenance of chronic pain.

2. Methods

2.1. Ethics

All experimental procedures were approved by the University of Sydney Human Research Ethics Committee and were consistent with the Declaration of Helsinki. Written informed consent was obtained from participants at the commencement of the study. Participants were also provided with an emergency buzzer while inside the MRI scanner so that they could stop the experiment at any time.

2.2. Participants

Thirty-seven healthy control participants were recruited for the study (15 male, 22 female, mean age 25.2 ± 1.0 [SEM] years, range 18.4–52.4 years). Inclusion criteria consisted of being over 18 years of age, able to provide written and informed consent, and no current or past diagnosis of any chronic pain condition or pain lasting longer than three months. Exclusion criteria consisted of the presence of any standard MRI-contraindications (current pregnancy, ferrous implants, etc.), or current or past chronic pain diagnosis. In order to evaluate the necessary number of participants required for this study, an *a priori* power analysis was performed using results from a previous imaging study investigating brainstem activation patterns in a within-subjects

paired design during placebo analgesia and hyperalgesia (Crawford et al., 2021). This revealed that a total sample size of 21 (7 per group) would be necessary to detect similar effect sizes with 95 % power ($d = 1.70$, $\alpha = 0.05$, power = 0.95). Before beginning the study, participants completed a data sheet recording demographic information including age, gender, and sex assigned at birth.

To explore psychological traits potentially influencing *offset* pain intensity changes, each participant completed a series of questionnaires used previously in pain processing and endogenous modulation studies: Behavioural inhibition and activation scales (BIS/BAS) (Carver and White, 1994), State-Trait Anxiety Inventory (STAI-S and STAI-T) (Spielberger et al., 1970), Pain Catastrophising Scale (PCS) (Sullivan et al., 1995) and the Life Orientation Test - Revised (LOT-R) (Goodin and Bulls, 2013).

2.3. Experimental calibration

Throughout the experiment, noxious stimuli were administered to the volar surface of the forearm using a 6.6×4 cm MRI-compatible Peltier-element thermode (QST labs, T09 stimulator, Strasbourg, France). During an initial calibration period, brief noxious stimuli were delivered using a pre-programmed series of temperatures. Each temperature stimulus lasted 14.4 s, including ramp-up and down periods (100° Celsius per second) and each stimulus was separated by a 14.4 s inter-stimulus-interval at a non-painful baseline temperature of 32° C. Participants rated their pain on-line using a horizontal 10 cm visual analogue scale (VAS) ranging between 0 and 100, where 0 was described as “no pain” and 100 as “the worst pain imaginable”. The temperature that elicited a moderate pain intensity (approximately 50/100) was used as the noxious stimulus temperature for the subsequent offset paradigm.

2.4. MRI data acquisition

Participants were positioned supine in a whole-body Siemens MAGNETOM 7 Tesla (7T) MRI system (Siemens Healthineers, Erlangen, Germany) with a combined single-channel transmit and 32-channel receive head coil (Nova Medical, Wilmington MA, USA). The thermode was attached to the volar surface of the right forearm using Velcro straps. A T1-weighted anatomical image set covering the whole brain was then acquired (repetition time=5000 ms, echo time=3.1 ms, raw voxel size= $0.73 \times 0.73 \times 0.73$ mm, 224 sagittal slices, scan time=7 mins). Following this, two fMRI acquisitions were performed, each consisting of a series of 403 gradient echo-planar measurements using blood oxygen level dependent (BOLD) contrast covering the entire brain. Images were acquired in an interleaved collection pattern with a multi-band factor of six and an acceleration factor of two (repetition time=800 ms, echo time=22.2 ms; raw voxel size= $1.6 \times 1.6 \times 1.6$ mm, 84 axial slices, scan time=6 min and 22.4 s). During the first fMRI series, following a 92.8 s baseline period, the thermode temperature was raised from 32° C to the temperature that evoked a moderate pain intensity. This moderate temperature was maintained for 14.4 s and then returned to 32° C for a period of 14.4 s. This “on-off” pattern was then repeated 7 times for a total of 8 moderate pain stimulation periods (Fig. 1a). During the second fMRI series, the thermode was repositioned to either a distal or proximal forearm location (counterbalanced between participants) relative to its first position, and an *offset* paradigm was performed. An “on” “off” paradigm similar to the first fMRI series was performed, however during the 14.4 s “on” period, after 4 s at the moderate pain intensity temperature (T1), the thermode temperature was raised by 1° C for 4 s (T2) and then lowered back to the moderate pain intensity temperature for 6.4 s (T3). The thermode temperature was then returned to 32° C for 14.4 s. Again, this pattern was repeated 7 times for a total of 8 “on” periods. During this scan, the participants continued to record their pain intensity ratings continuously. Participants used an MR-compatible button box to continuously report their pain perception during both fMRI scans. The VAS scale was shown on a reflected digital

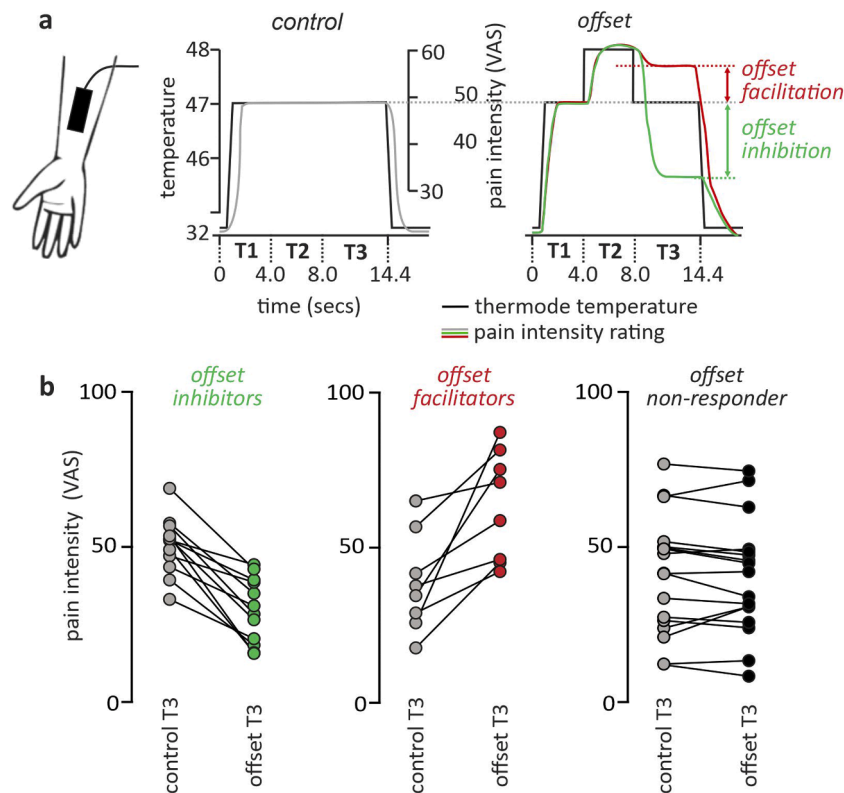


Fig. 1. Experimental protocol and pain intensity changes. A) Offset paradigm. An illustration of the method used to delineate *offset inhibitor* (OI); *offset facilitator* (OF); and non-responder groups. During the control scan, thermode temperatures were raised and maintained at a set temperature that would elicit a moderate pain intensity. During the *offset* scan, the thermode temperature was initially raised to the same level and held for 4 s (T1), then increased by 1 °C and held for 4 s (T2), then decreased by 1 °C for 6.4 s (T3), followed by a reduction back to baseline levels (32 °C). Individual pain intensity differences of T3 periods during the control and *offset* scans were then calculated and used to categorize participants into one of the three groups. Individuals displaying significant pain decreases were placed into the OI group, those with pain increases into the OF group, and the remaining participants into the non-responder group. B) Pain intensity changes. Plots of individual mean (averaged over 8 stimulus periods) T3 period pain intensity ratings during control (grey circles) and *offset* (green, red, and black circles) scans separated into OI, OF and non-responder groups. Note that OI participants display pain decreases, OF pain increases and non-responders little to no pain intensity change, during *offset* compared with control scans.

screen at the end of the magnet bore, and participants controlled the position of a slider to report their pain continuously by holding the left (moved slider towards 0) or right (moved slider towards 100) buttons with their left middle and index fingers. The sampling rate for the digital VAS system was 0.5 s; resulting in eight data points corresponding to each T1 and T2 period; and thirteen data points to T3. After both fMRI series were recorded, a reverse-phase encoded image set was acquired consisting of 10 measurements in the inverse field orientation, enabling distortion correction through TopUp software (repetition time=800 ms, echo time=22.2 ms; raw voxel size = $1.6 \times 1.6 \times 1.6$ mm, 84 axial slices, scan time=8 sec).

2.5. MRI preprocessing

MRI images were processed using Statistical Parametric Mapping version 12 (SPM12). Images were distortion corrected using TopUp (Andersson et al., 2003; Smith et al., 2004), slice-time and movement corrected. Movement parameters were inspected to ensure that each participant had no >1 mm linear or 0.5° rotational movement. The linear detrending function of SPM12 was applied to resulting images to model and remove linear drifts in signal across time at a voxel-wise level, and physiological noise relating to cardiac (60–120 beats per minute $\pm$ 1 harmonic) and respiratory (8–25 breaths per minute $\pm$ 1 harmonic) frequency and harmonics were modelled and subsequently filtered using the DRIFTER toolbox (Sarkka et al., 2012). Temporal signal to noise ratio (tSNR) images were calculated prior to and following denoising to ensure a significant increase in tSNR was

observed at both the wholebrain level, and within discrete brainstem nuclei relevant to endogenous pain modulation (i.e. the left and right locus coeruleus (Supplementary Table 1). The 6-parameter movement-related signal changes were modelled and removed using a linear modelling of realignment parameters (LMRP) procedure. Each individual's fMRI image sets were then co-registered to their own T1-weighted anatomical, the T1-weighted image then spatially normalized to the IXI555 template in Montreal Neurological Institute (MNI) space, with these parameters applied to fMRI image sets. In these images, the brainstem and cerebellum were isolated using the spatially unbiased infra-tentorial template (SUIT) toolbox (Diedrichsen, 2006), for both the fMRI and T1 image sets, and then normalized to the brainstem- and cerebellum-only SUIT template in MNI space. The resulting brainstem fMRI images were resliced to 0.5 mm isotropic voxel size and spatially smoothed using a 1 mm FWHM Gaussian filter.

2.6. Psychophysics analysis

One-way ANOVA analyses were first conducted across the whole participant cohort to determine behavioural offset analgesia using both pain intensity ratings recorded between the control scan T3 and offset scan T3 periods, as well as between offset scan T1 and T3 periods. Next, using a bootstrapped permutation procedure, participants were categorized into one of three groups: i) *offset inhibitors* (OI): T3 pain intensity was significantly lower during *offset* compared with control scan; ii) *offset facilitators* (OF): T3 pain intensity was significantly higher during *offset* compared with control scan; or iii) non-responders: T3 pain

intensity was not significantly different during *offset* compared with control scan. Briefly, mean VAS ratings consistent with volumes of the corresponding T3 period of each of the 8 noxious stimuli applied during the control series were entered into a permutation model, where 10,000 artificial samples were generated with replacement. This artificial sample was significance tested to 10,000 artificial samples generated from the VAS ratings of the T3 period of the 8 noxious stimuli delivered during the *offset* scan series. If the mean difference between the two series was significant, a participant was placed into either: the offset inhibitor group if pain intensity was significantly lower during the *offset* scan T3 periods, or the offset facilitator group if the pain was greater during the *offset* scan T3 periods compared with the control scan T3 periods. Those individuals that displayed no significant pain intensity differences were classified as non-responders. One-way ANOVA ($p < 0.05$) analyses were also conducted to determine if there were differences in the temperature applied or pain intensity ratings reported between all three groups during the control series T3, and offset series T1, T2, and T3 periods, to ensure that any of the reported pain effects were unrelated to baseline thermal sensitivity. Main effect of sex analyses were also conducted to ensure sex assigned at birth did not confound behavioural offset analgesia.

2.7. fMRI statistical analysis

To determine significant changes in signal intensity associated with offset inhibitory or paradoxical facilitatory responses, a repeating boxcar model convolved with a canonical hemodynamic response function was applied to each of the fMRI series. Within this model, dynamic volumes overlying T3 periods were assigned a value of 1, and all other scan volumes were assigned a value of 0. The contrast images generated for each control and *offset* functional image series were then placed into a random effects second level, between group analysis to determine significant differences in brainstem signal intensity changes between the OI and non-responder groups, and between the OF and non-responder groups. Following an initial significance threshold of $p < 0.001$ uncorrected for multiple comparisons, with a cluster extend threshold of 10 contiguous voxels, clusters underwent small volume correction using individual region-of-interest masks derived from the brainstem atlas of Coulombe and colleagues (Coulombe et al., 2021), ensuring cluster peaks survived correction at a family-wise error rate of $p < 0.05$ to reduce the likelihood of Type 1 errors.

For the OI versus non-responder group analysis, signal intensity changes (β -values) of each significant cluster were extracted from the brainstem contrast maps for control and *offset* scans in each participant, and individual values plotted. For each cluster, β -values were extracted from each of the three groups (OI, OF, non-responders), mean ($\pm$ SEM) values tabulated and significant differences between OF and non-responder groups determined using paired t -tests ($p < 0.05$). Significant differences between OI and non-responders were not determined post-hoc to avoid double-dipping. In addition, for each cluster, percentage signal intensity changes relative to the baseline period were calculated for both the control and *offset* scans, and mean ($\pm$ SEM) signal intensity change, differences (offset - control) for T1, T2, and T3 periods plotted for each of the 8 stimulus periods. In addition, the mean ($\pm$ SEM) signal intensity change, differences (offset - control) for T1, T2 and T3 periods averaged over the 8 stimulus periods were plotted and significant differences between each of the three periods of noxious stimulation (T1, T2, T3) determined using paired t -tests ($p < 0.05$). The same procedures were then repeated for the OF versus non-responder group analysis. Additionally, difference maps encoding signal intensity change between the offset and control series were generated for all participants (offset minus control), with beta-values extracted for each cluster resulting from either OI- or OF versus non-responder group analysis. These difference map values encoding the change in signal intensity between the control and offset functional series were investigated for associations with the magnitude of offset analgesia generated (control

T3 VAS average minus offset T3 average VAS) across our entire sample through linear regression analysis. All significant clusters were overlaid onto a mean brainstem anatomical image set for visualization purposes. For brainstem cluster localization and identification, we used the atlas by Coulombe and colleagues (Supplementary Table 2) (Coulombe et al., 2021).

3. Results

3.1. Psychophysics

Initial assessment of pain intensity ratings across all 37 participants revealed no significant behavioural effect of offset analgesia either between T3 periods of the control and offset scans (VAS mean $\pm$ SEM: control 43.59 ± 2.61 ; offset 41.89 ± 3.22 ; $p = 0.60$), nor between T1 and T3 periods of the offset scan (VAS mean $\pm$ SEM: offset T1 38.24 ± 2.53 ; offset T3 41.89 ± 3.22 ; $p = 0.10$) (Supplementary Fig. 1; Supplementary Table 3). However, permutation testing applied to pain intensity changes during T3 periods of the control and offset scans revealed 12 participants as showing a significant reduction in pain, being allocated to the OI group (VAS mean $\pm$ SEM: control 50.59 ± 2.55 ; offset 29.81 ± 2.89 ; $p < 0.001$). Dissimilarly, 8 participants demonstrated a significant increase in pain, being allocated to the OF group (control 38.36 ± 5.23 ; offset 63.23 ± 5.84 ; $p = 0.007$), and in 17 participants no change in pain was recorded, with these participants considered non-responders (control 41.12 ± 4.47 ; offset 40.37 ± 4.33 ; $p = 0.50$) (Fig. 1b / Table 1). These relationships in pain intensity change within groups remained consistent when considering changes between offset T1 and T3 periods, with exception to non-responders who demonstrated pain increases during T3 relative to T1 periods (Table 1). While the mean pain intensity rating during T1 periods of the offset series were significantly different between groups ($p = 0.04$), there were no significant differences in pain intensity during either control T3 or Offset T2 periods, nor in calibrated thermode temperature between the three groups (Table 1). Moreover, single factor ANOVA revealed no main effect of group in thermode ramp perception – that is, the change in pain between T1 and T2 periods of the offset protocol – was present (Mean $\pm$ SEM VAS change: OI = 17.21 ± 0.75 ; non-responder = 18.88 ± 0.39 ; OF = 18.66 ± 1.10 ; $p = 0.34$). In addition, no significant differences in pain intensity ratings during any of the control T3; or Offset T1, T2, or T3 periods was identified between sexes assigned at birth; and no main effects of sex were identified in pain intensity changes between control T3 and offset T3, nor offset T1 and offset T3 pain periods (Supplementary Table 3). Finally, comparison of questionnaire scores between the groups did not reveal any significant differences for BIS, BAS, STAI-S, STAI-T, and PCS, although LOT-R was significantly higher in OI compared with the non-responder group at an exploratory level without applying correction for multiple administered questionnaires (all $p > 0.05$; Table 2).

3.2. Offset inhibitor group evoked brainstem signal intensity changes

Analysis of significant differences between control and offset scan T3 periods between OI and non-responder groups revealed that T3 pain intensity reductions during the control compared with offset scans were associated with signal intensity change, decreases and increases in multiple brainstem regions. Signal intensity decreases occurred in the ventral tegmental area, left and right locus coeruleus (LC), left dorso-lateral medulla and the caudal nucleus tractus solitarius, whereas signal intensity increases occurred in the left parabrachial nucleus, right sub-coeruleus nucleus and right substantia nigra (Fig. 2a, Table 3). Extraction of β -values revealed that for each cluster, OI participants displayed consistent increased or decreased signal intensity changes whereas, non-responders showed a range of control versus offset scan signal intensity changes (Fig. 2b). Extraction of β -values from all three groups revealed that these significant signal changes were largely restricted to the OI group. That is, the OF group did not display significant T3 signal

Table 1
Pain intensity ratings during control T3 and offset T1, T2, and T3 periods for non-responder, *offset inhibitor*, and *offset facilitator* groups.

	Control T3	Offset T1	Offset T2	Offset T3	Control T3 vs Offset T3 p-value	Offset T1 vs Offset T3 p-value
non-responders	41.12±4.47	32.33±3.23	51.21±4.45	40.37±4.33	0.50	0.005
offset inhibitors	50.59±2.55	39.54±3.53	56.75±3.92	29.81±2.89	<0.001	<0.001
offset facilitators	38.36±5.23	48.86±6.15	67.52±6.02	63.23±5.84	0.007	0.002
p-value	0.18	0.04	0.11	<0.001		

All values are expressed as mean ±SEM. P-values are produced from one-factor ANOVA either within- or between groups.

Table 2
Thermode temperature and questionnaire values for non-responder, *offset inhibitor* and *offset facilitator* groups.

	Thermode temperature (°Celsius)	BIS	BAS	SAI	TAI	PCS	LOT-R
non-responders	46.71±0.15	22.69±0.72	38.38±0.97	31.59±2.34	21.94±2.43	12.13±1.46	13.31±1.20
offset inhibitors	46.46±0.26	21.25±1.30	39.42±1.58	32.50±2.41	24.25±2.76	11.25±1.74	17.00±1.24
offset facilitators	46.38±0.13	19.88±1.19	41.13±1.32	31.75±2.42	20.25±2.27	9.38±3.28	17.25±1.26
p-value	0.47	0.21	0.39	0.96	0.63	0.65	0.05*

All values are expressed as mean ±SEM. P-values are produced from one-factor ANOVA between groups. Note that only the life orientation test – revised displays between-group significance. BIS: behavioural inhibition scale; BAS: behavioural activation scale; PCS: pain catastrophising scale; SAI: state anxiety inventory; TAI: trait anxiety inventory; LOT-R: life orientation test – revised.

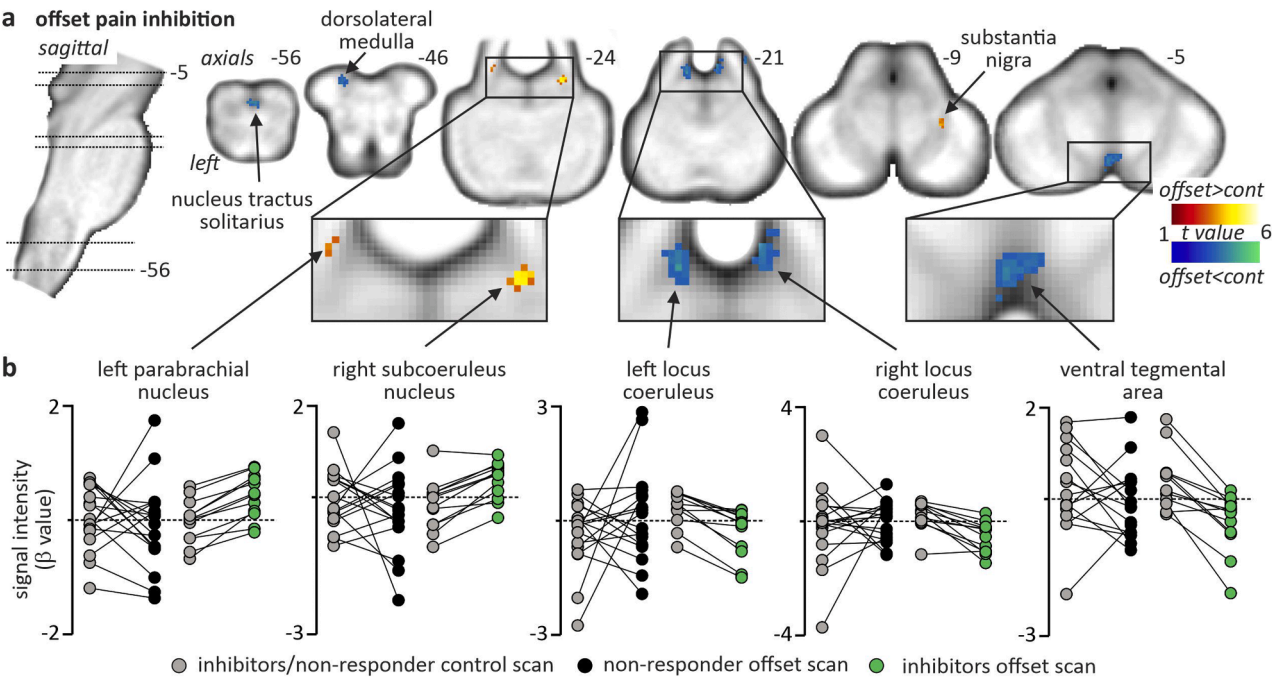


Fig. 2. Brainstem signal intensity changes associated with offset pain decreases. A) Clusters displaying significant signal intensity increases (hot colour scale) or decreases (cool colour scale) in the offset inhibitory compared with non-responder groups. Clusters are overlaid onto a mean T1-weighted anatomical image set. Slice locations in Montreal Neurological Institute space are indicated at the top right of each slice. B) Plots of signal intensity changes during the control (grey circles) and offset scans for non-responder (black circle) and offset inhibitor (green circles) participants. Note that offset inhibitors show consistent signal intensity increases or decreases whereas non-responders show variable responses.

intensity change differences, apart from the dorsolateral medulla which displayed opposing changes in OF compared with OI groups (Table 4). Moreover, extraction of difference map values revealed a significant inverse relationship between signal intensity change during T3 periods and offset analgesia magnitude in all participants within specifically the right locus coeruleus, the ventral tegmental area, and the left dorsolateral medulla. That is, the greater the magnitude of behavioural offset analgesia, the greater the decrease in signal within these regions during the offset relative to the control scan T3 periods (Supplementary Table 4).

Extraction of percentage signal intensity change, differences (offset-control) for T1, T2, and T3 periods revealed that for all clusters, apart from the substantia nigra, that significantly greater signal changes were

present during T3 periods when comparing offset versus control scans, and that signal intensity increases were larger during T3 compared with T1 periods, and vice versa for signal decreases (Fig. 3a, Table 5). Subtraction of signal intensity change, differences (offset-control) between T3 and T1 periods (T3-T1) revealed that signal changes were consistent across the 8 noxious stimuli, with at least 6 of the 8 noxious stimuli showing similar signal change directions (Fig. 3b).

3.3. Offset facilitator group evoked brainstem signal intensity changes

In striking contrast to the number of brainstem regions displaying altered signal intensity changes during offset inhibition, offset facilitation was associated with only two brainstem clusters with significant

Table 3

Location in Montreal Neurological Institute Space (MNI), cluster size and *t* values of clusters that displayed significant signal intensity change differences during T3 period between control and *offset* scans.

region	MNI coordinates			cluster size	t-value
	X	Y	Z		
<i>Inhibitors: offset different to control scan</i>					
<i>offset>control</i>					
left parabrachial nucleus	−5.5	−38.5	−26	33	5.46
right subcoeruleus nucleus	8	−37.5	−24	22	5.65
<i>offset<control</i>					
left locus coeruleus	−2.5	−38	−21	46	5.58
right locus coeruleus	3.5	−39.5	−21.5	34	4.98
ventral tegmental area	0	−14.5	−4.5	130	4.97
left dorsolateral medulla	−4.5	−42.5	−45.5	37	4.49
nucleus tractus solitarius	1.5	−44	−55.5	19	4.99
right substantia nigra	10	−23.5	−8.5	17	4.69
<i>Facilitators: offset different to control scan</i>					
<i>offset>control</i>					
pontine reticular nucleus	7	−35	−34	38	6.37
<i>offset<control</i>					
rostral ventromedial medulla	1	−25	−42	26	6.36

signal intensity changes. Analysis of significant differences between control and *offset* scan T3 periods between OF and non-responder groups revealed that T3 pain intensity increases during the *offset* versus control scans were associated with signal intensity decreases in the rostral ventromedial medulla (RVM), and signal increases in the pontine reticular nucleus (Fig. 4a, Table 3). Extraction of β -values revealed that for both clusters, OF participants consistently displayed signal intensity change, decreases or increases whereas, non-responders showed a range of control versus *offset* scan signal intensity changes (Fig. 4b). Extraction of β -values from all three groups revealed that the significant signal changes were restricted to the OF group, that is, the OI group did not display significant T3 signal intensity change differences (Table 4). Extraction of difference map values revealed that neither the pontine reticular nor the rostral ventromedial medulla displayed a linear association between the magnitude of behavioural offset analgesia and change in signal intensity between offset and control T3 periods (Supplementary Table 4).

Extraction of percentage signal intensity change, differences (*offset*-control) for T1, T2 and T3 periods revealed that only the RVM displayed significantly lower signal intensity changes during T3 periods compared with T1 periods (Fig. 5a, Table 5). Subtraction of these signal intensity change, differences (*offset*-control) between T3 and T1 periods (T3-T1)

revealed that signal intensity change, differences for the RVM were relatively consistent, with 6 of the 8 noxious stimuli showing signal change decreases (Fig. 5b).

4. Discussion

Our data show that an offset analgesia paradigm results in 32 % of individuals displaying disproportionate pain intensity decreases, 22 % displaying disproportionate pain intensity increases and the remaining 46 % displaying no significant pain intensity change when compared to a control stimulus paradigm. Whilst at a cohort level we identified no significant main effect on pain intensity, permutation testing revealed our paradigm induced clear behavioural effects within subgroups consistent with offset analgesia and paradoxical hyperalgesia. There were no significant associations between *offset* pain decreases or increases with measures of behavioural inhibition and activation, anxiety, or pain catastrophising. However, we did identify a weak association with trait optimism. Our brainstem fMRI analysis revealed that *offset* pain inhibition is associated with robust signal changes in a number of brainstem regions including the locus coeruleus, the subcoeruleus and the parabrachial nucleus. In striking contrast, *offset* pain facilitation was associated with signal changes in only two brainstem regions, the pontine reticular nucleus and the RVM. Of note is that only a select few of the brainstem regions identified demonstrated significant linear relationships between signal intensity change and behavioural analgesia across our entire sample cohort – suggesting that regional associations identified largely reflect unique functional engagement of brainstem systems dependent on the behavioural profile as showing an inhibitory, non-, or facilitatory offset response. Under our paradigm, no significant signal intensity changes were observed in alternate brainstem pain modulatory sites, such as the midbrain periaqueductal gray matter, the subnucleus reticularis dorsalis, or in the A5 cell group.

We have reported previously, that the *offset* paradigm employed in this study evokes similar proportions of people that show pain intensity decreases, increases, and no responses, to those evoked during placebo and conditioned pain modulation paradigms (L.S. Crawford et al., 2025). In addition, the lack of significant associations between either *offset* pain decreases or increases, with any of the psychological measures evaluated including behavioural inhibition and activation, anxiety, and pain catastrophising suggests strongly that *offset* related pain intensity changes are not influenced by psychological state or factors. A suggestion consistent with the idea that *offset* pain changes are largely sensory-driven modulation of pain phenomena (Martucci et al., 2012).

Table 4

Signal intensity changes (mean $\pm$ SEM β values) for significant clusters in non-responder, inhibitor and facilitator groups during T3 period for the control and *offset* scans.

	Non-responders			Inhibitors			Facilitators		
	control	offset	p-value	control	offset	p-value	control	offset	p-value
Inhibitors: offset different to control scan									
offset>control									
left parabrachial nucleus	0.05, $\pm$ 0.48	-0.10, $\pm$ 0.22	$p = 0.48$	-0.02, $\pm$ 0.11	0.43, $\pm$ 0.11	$p < 0.05^*$	0.01, $\pm$ 0.11	0.21, $\pm$ 0.16	$p = 0.24$
right subcoeruleus nucleus	-0.05, $\pm$ 0.18	-0.26, $\pm$ 0.26	$p = 0.50$	-0.15, $\pm$ 0.15	0.40, $\pm$ 0.11	$p < 0.05^*$	-0.21, $\pm$ 0.16	-0.09, $\pm$ 0.14	$p = 0.59$
offset<control									
left locus coeruleus	-0.38, $\pm$ 0.30	0.23, $\pm$ 0.42	$p = 0.25$	0.34, $\pm$ 0.13	-0.31, $\pm$ 0.18	$p < 0.05^*$	0.60, $\pm$ 0.28	0.23, $\pm$ 0.29	$p = 0.10$
right locus coeruleus	-0.14, $\pm$ 0.39	0.01, $\pm$ 0.20	$p = 0.70$	0.24, $\pm$ 0.14	-0.56, $\pm$ 0.15	$p < 0.05^*$	0.59, $\pm$ 0.27	0.08, $\pm$ 0.37	$p = 0.29$
ventral tegmental area	0.22, $\pm$ 0.28	-0.07, $\pm$ 0.23	$p = 0.31$	0.40, $\pm$ 0.18	-0.54, $\pm$ 0.18	$p < 0.05^*$	0.22, $\pm$ 0.18	0.04, $\pm$ 0.17	$p = 0.45$
left dorsolateral medulla	0.01, $\pm$ 0.16	0.03, $\pm$ 0.23	$p = 0.91$	0.33, $\pm$ 0.15	-0.33, $\pm$ 0.13	$p < 0.05^*$	-0.05, $\pm$ 0.05	0.12, $\pm$ 0.06	$p = 0.03^{\#}$
nucleus tractus solitarius	0.14, $\pm$ 0.18	0.15, $\pm$ 0.28	$p = 0.98$	0.33, $\pm$ 0.14	-0.36, $\pm$ 0.14	$p < 0.05^*$	0.08, $\pm$ 0.17	0.20, $\pm$ 0.15	$p = 0.75$
right substantia nigra	0.15, $\pm$ 0.15	0.18, $\pm$ 0.26	$p = 0.91$	0.13, $\pm$ 0.08	0.71, $\pm$ 0.10	$p < 0.05^*$	-0.08, $\pm$ 0.21	0.41, $\pm$ 0.21	$p = 0.09$
Facilitators: offset different to control scan									
offset>control									
pontine reticular nucleus	0.23, $\pm$ 0.19	-0.13, $\pm$ 0.27	$p = 0.30$	0.11, $\pm$ 0.17	0.35, $\pm$ 0.34	$p = 0.45$	-0.36, $\pm$ 0.15	0.94, $\pm$ 0.12	$p < 0.05^*$
offset<control									
rostral ventromedial medulla	-0.03, $\pm$ 0.17	-0.36, $\pm$ 0.14	$p = 0.12$	0.14, $\pm$ 0.15	0.03, $\pm$ 0.07	$p = 0.44$	0.08, $\pm$ 0.09	-0.59, $\pm$ 0.17	$p < 0.05^*$

* voxel-by-voxel analysis.

post-hoc paired *t*-test analysis.

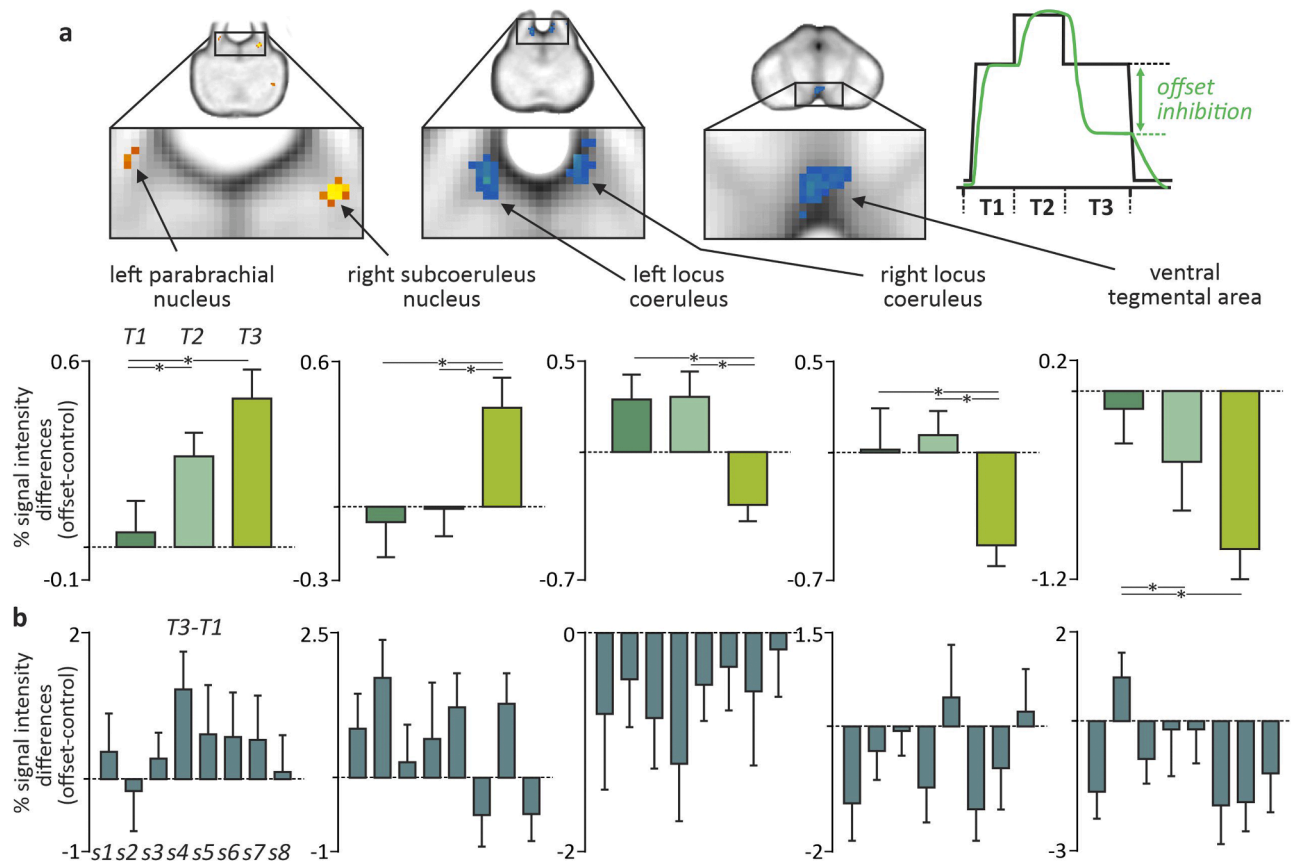


Fig. 3. Percentage signal intensity changes associated with *offset* pain decreases. A) Plots of percentage (relative to baseline) signal intensity changes during *offset* relative to control scans (*offset-control*) during T1, T2 and T3 periods. It is clear that for areas displaying signal intensity decreases such as the locus coeruleus and ventral tegmental area, T3 signal intensity change, differences were significantly reduced during the *offset* scan and were also significantly lower than signal intensity change, differences during T2 and T1 periods, and vice versa for T3 signal increases. * $p < 0.05$ paired t -tests. B) Plots of percentage T3-T1 signal intensity change differences (*offset-control*) for each of the 8 stimulation periods. Note the consistency in response patterns whereby at least 6 of the 8 stimulus periods displayed similar changes. s1 - s8: stimulus 1 – stimulus 8.

Table 5

Signal intensity changes (mean $\pm$ SEM % change) for significant clusters in non-responder, inhibitor and facilitator groups during T1, T2 and T3 period for the control and *offset* scans.

Inhibitors: <i>offset</i> - control scan	T1	T2	T3
<i>offset</i>>control			
left parabrachial nucleus	0.05 $\pm$ 0.10	0.29 $\pm$ 0.08 [†]	0.47 $\pm$ 0.09*
right subcoeruleus nucleus	-0.07 $\pm$ 0.14	0.003 $\pm$ 0.13	0.41 $\pm$ 0.12*,#
<i>offset</i><control			
left locus coeruleus	0.30 $\pm$ 0.14	0.32 $\pm$ 0.13	-0.29 $\pm$ 0.09*,#
right locus coeruleus	0.01 $\pm$ 0.23	0.09 $\pm$ 0.13	-0.51 $\pm$ 0.11*,#
ventral tegmental area	-0.11 $\pm$ 0.22	-0.45 $\pm$ 0.31 [†]	-1.01 $\pm$ 0.18*
left dorsolateral medulla	-0.12 $\pm$ 0.15	-0.11 $\pm$ 0.16	-0.55 $\pm$ 0.12*,#
nucleus tractus solitarius	0.06 $\pm$ 0.10	0.05 $\pm$ 0.07	-0.51 $\pm$ 0.15*,#
right substantia nigra	0.08 $\pm$ 0.20	0.19 $\pm$ 0.21	0.50 $\pm$ 0.11
Facilitators: <i>offset</i> - control scan			
<i>offset</i>>control			
pontine reticular nucleus	0.86 $\pm$ 0.40	0.71 $\pm$ 0.40	1.32 $\pm$ 0.29
<i>offset</i><control			
rostral ventromedial medulla	0.91 $\pm$ 0.46	0.65 $\pm$ 0.41	-0.80 $\pm$ 0.33*,#

* T3 different T1.

T3 different T2.

[†] T2 different T1; $p < 0.05$ post-hoc paired t -test analysis.

However, we did find significantly higher LOT-R scores in the OI group when compared with the non-responder group, suggesting that greater optimism is associated with greater pain inhibition, a relationship we reported previously for placebo-related pain inhibition (L.S. Crawford et al., 2025).

One of the most robust signal intensity changes during *offset* pain reductions occurred bilaterally in the locus coeruleus. These signal changes were observed only in the OI group and never in either the non-responders or OF groups. We hypothesised originally, that offset analgesia would be driven by changes in the PAG-RVM-dorsal horn/SpV pathway, since this is the canonical brainstem analgesic circuit. However, the LC is also known to be able to powerfully modulate incoming noxious information. The LC is a major source of noradrenaline which can inhibit or facilitate neurotransmission via activation of α -1 and α -2 adrenoreceptors (Llorca-Torralba et al., 2016; Wei and Pertovaara, 2006). As the primary site of noradrenaline release, the LC forms connection pathways with several cortical areas responsible for pain processing and cognitive evaluation. Specifically, during the expression of attentional analgesia, the LC forms a bidirectional loop with the anterior cingulate cortex, and demonstrates altered coupling to the RVM as a function of both attentional demand and noxious intensity in this endogenous pain modulatory paradigm (Oliva et al., 2022; Oliva et al., 2021; Taylor and Westlund, 2017). Moreover, the LC directly innervates the dorsolateral and medial prefrontal cortices, two core cortical areas in establishing top-down pathways for pain modulation (Dave et al., 2025). Spinally, noradrenaline released from LC projections to the dorsal horn/SpV can inhibit nociceptive neurotransmission (Jones and

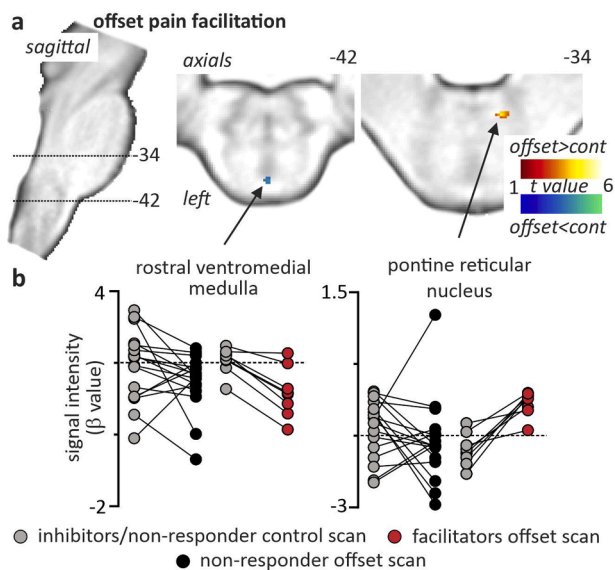


Fig. 4. Brainstem signal intensity changes associated with *offset* pain increases. A) Clusters displaying significant signal intensity increases (hot colour scale) or decreases (cool colour scale) in the *offset* facilitatory compared with non-responder groups. Clusters are overlaid onto a mean T1-weighted anatomical image set. Slice locations in Montreal Neurological Institute space are indicated at the top right of each slice. B) Plots of signal intensity changes during the control (grey circles) and *offset* scans for non-responder (black circle) and *offset* facilitators (red circles) participants. Note that *offset* facilitators show consistent signal intensity increases or decreases whereas non-responders show variable responses.

Gebhart, 1986; Tsuruoka et al., 2003), and intrathecal administration of alpha-2 adrenergic agonists reduces capsaicin-induced pain and hyperalgesia in healthy individuals (Eisenach et al., 1998). Functional MRI studies have shown that activity in both the LC and dorsal horn are inversely correlated with inter-individual pain intensity ratings during noxious thermal stimulation (Khan and Stroman, 2015), and several studies have begun to explore the role of LC in pain modulation. For example, acupuncture analgesia is associated with LC signal decreases (Napadow et al., 2009), and using ultra-high field fMRI we have shown that placebo analgesic responses are associated with signal decreases in the LC (Crawford et al., 2021). An earlier fMRI study suggested that *offset* analgesia involved signal changes in the PAG-RVM circuit, however the signal changes illustrated in that study encompassed the LC. On balance, these earlier findings and the data we present here are consistent with the conclusion that the LC plays an important role in human pain modulation paradigms (Derbyshire and Osborn, 2009). Our data are consistent with *offset* pain changes being primarily associated with LC functional changes – with the right LC specifically being one of only three sites showing both a group-level change in *offset* inhibitors as well as a linear association between the magnitude of *offset* analgesia elicited across all participants. These data also support the functional relevance in *offset* analgesia of several other brainstem nuclei with known roles in pain processing and modulation. Whilst our findings do not preclude involvement of the canonical PAG-RVM circuit as being associated with the *offset* analgesia phenomenon, they support that in this specific paradigm and method of data processing that no subthreshold effects are observed within the PAG.

In addition to the LC, we also report signal intensity decreases during *offset* pain intensity decreases in the dorsolateral medulla and in the NTS, as well as signal increases in the subcoeruleus nucleus. It has been shown that acute pain is associated with changes in autonomic function, including altered heart rate, blood pressure, and sympathetic activity (A. R. Burton et al., 2009; A.R. Burton et al., 2009; Kyle and McNeil, 2014), and it is well established that the NTS is a critical region regulating

autonomic function and mediating autonomic changes (Dampney, 1994; Macefield and Henderson, 2010). The NTS, and in particular the caudal NTS, receives both baroreceptor and chemoreceptor inputs and modulates activity in the rostral ventrolateral medulla (RVLM), the region containing premotor sympathetic neurons (Dampney, 1994). Almost all changes in sympathetic activity are mediated by the RVLM, and while we know of no study that has measured autonomic changes during *offset* analgesia, it has been reported that baseline heart rate variability is associated with the magnitude of *offset* analgesia responses (Kyle and McNeil, 2014). It is possible that *offset* pain intensity changes are associated with altered autonomic responses, and that such changes are reflected by the observed activity changes in nodes of the NTS-RVLM pathway. In addition, the LC has also been shown to alter autonomic function, with direct activation of LC neurons using intracranial microinjections of glutamate evoking increases in heart rate and blood pressure, via its projections to the RVLM and other brainstem autonomic nuclei (Chen and Chai, 2002; Van Bockstaele et al., 1989). The subcoeruleus nucleus is also involved in autonomic control with direct electrical stimulation of this region evoking both heart rate and blood pressure increases (Gurtu et al., 1984), interestingly this region is also critical for the transition to REM sleep (Lu et al., 2006).

In striking contrast to the brainstem regions displaying altered signal changes during *offset* pain reductions, pain increases were not associated with significant changes in brainstem signals. The notable exception was the RVM, which showed significantly different signal intensity changes during T3 in the OF group compared with controls. While it is well-known that the RVM projects to both the dorsal horn of spinal cord and the spinal trigeminal nucleus to modulate pain, it is generally thought that these projections are under significant regulation by inputs from the PAG and parabrachial nucleus (Chen and Heinricher, 2022). However, it has recently been shown that cerebral cortical regions including the sensorimotor cortex sends direct projections to the RVM, which are likely the pathway by which sensorimotor cortex stimulation reduces chronic pain (Garcia-Larrea, 2023; Hamani et al., 2021; Wang et al., 2025). Since we observe significant signal intensity change within the RVM, yet no subthreshold PAG-related effects as associated with *offset* pain increases, it is likely a separate functional system underpinning these responses. Further studies exploring these connections with brainstem pain modulatory nuclei such as the RVM during *offset* pain facilitation through psychophysiological interaction or dynamic causal modelling analyses would address this knowledge gap.

Although we investigated pain-free individuals in this study, there is growing clinical evidence that *offset* analgesia responses are altered in individuals with chronic pain (Shulman et al., 2020; Cosentino et al., 2025; Niesters et al., 2011). Furthermore, there is preclinical evidence that altered LC function is involved in the development and/or maintenance of chronic neuropathic pain. For example, some investigations suggest that in neuropathic pain, the LC inhibits nociception at the dorsal horn leading to enhanced mechanical sensitivity (Hayashida et al., 2012; Jasmin et al., 2003), whereas other studies suggest a pro-nociceptive role of the LC as contributing to the maintenance of sensory and affective behavioural responses (Boorman et al., 2019; Kaushal et al., 2016). These preclinical studies suggest that neuroplastic changes within the LC contribute to pain development and it is possible that such changes relate to the altered *offset* analgesic responses reported in individuals with chronic pain.

5. Limitations and future directions

We note some limitations. Firstly, our method for triggering *offset* analgesia utilizes 14.4 s stimulus blocks, with a T2 period of 4 s, and a T3 period of 6.4 s, which are relatively shorter than commonly used outside the MRI scanner environment. Indeed, it has been shown that behavioural *offset* analgesia increases as a function of longer T3 time periods (Asplund et al., 2021), and the reduced proportion of *offset* inhibition we observe here may be attributed to these shorter stimulus durations.

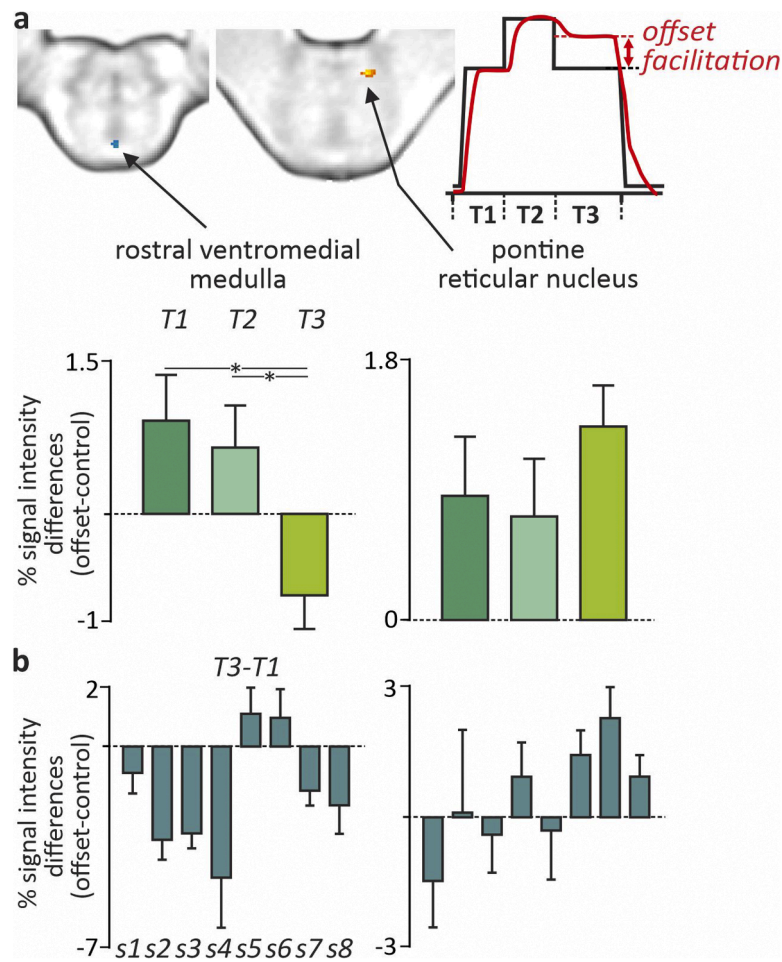


Fig. 5. Percentage signal intensity changes associated with *offset* pain increases. A) Plots of percentage (relative to baseline) signal intensity changes during *offset* relative to control scans (*offset-control*) during T1, T2 and T3 periods. It is clear that for the rostral ventromedial medulla, T3 signal intensity change differences were significantly reduced during the *offset* scan and were also significantly lower than signal intensity change, differences during T2 and T1 periods. In contrast, for the pontine reticular nucleus, whilst T3 signal intensity changes were greater during *offset* relative to control, they were not significantly different to T2 or T1 periods. * $p < 0.05$ paired t -tests. B) Plots of percentage T3-T1 signal intensity change differences (*offset-control*) for each of the 8 stimulation periods. Note the consistency in response patterns for the rostral ventromedial medulla whereby 6 of the 8 stimulus periods displayed similar changes. s1 - s8: stimulus 1 – stimulus 8.

Further, since haemodynamic delay occurs in the order of seconds, it is likely that the functional changes we capture with these shorted T3 periods of our design reflect immediate changes in brainstem systems resulting from the expression of offset analgesia. Since the function of certain brainstem pain modulatory regions such as the PAG and pontine nuclei are known to interact with noxious stimulus duration (Petrovic et al., 2004), whether the same regions we identify remain involved or indeed if alternate regions become more critical for the maintenance of offset analgesia over longer stimulus time periods is of interest. Additionally, we characterized offset analgesia as the magnitude of pain intensity change between two separate pain conditions (i.e. control and offset T3 periods), where typically it is evaluated as changes in pain between T1 and T3 periods of the offset condition alone. However, the distribution of responses we detect with this method were consistent with alternate pain modulatory mechanisms such as placebo analgesia and conditioned pain modulation, and pain intensity change directions were similar, and significance retained when analysing values between offset T1 and T3 periods, suggesting our method does not deviate from the traditional approach. Moreover, this method enabled us to conduct paired analyses at a voxel-wise brain imaging level, and answer research questions surrounding differences in brainstem activation patterns between typical pain perception events and during offset inhibitory and facilitatory responses. An alternate approach using T1 and T3 offset periods to characterize offset analgesia magnitude and including a trace

of pain intensity ratings as a covariate in functional imaging analysis throughout the entire offset stimulus block could be employed, which may better resolve the change in brainstem function resulting from the entire protocol, rather than what we demonstrate as changes in brainstem activity between acute pain encoding and during an offset analgesia protocol. Similarly, we elected to group participants as OI, OF and non-responders, which was important for evaluating brainstem circuitry involved during offset pain decreases and increases, however this reduced the overall power of the study, and findings especially in the OF group ($n = 8$), should be considered with respect to these small group sizes. Future investigations with expanded study cohorts are recommended to ensure the functional systems we identify remain consistent with more robust group sizes. Secondly, we employed an initial threshold of $p < 0.001$ uncorrected for multiple comparisons, with a cluster extent threshold of 10 contiguous voxels. To reduce the potential for type 1 errors we performed small volume correction using brainstem co-ordinates of the Coulombe atlas, and furthermore, our results were largely limited to brainstem nuclei previously shown to play a critical role in modulating pain. Finally, while the enhanced spatial resolution provided by ultra-high field imaging allowed us to describe voxel peaks within anatomically meaningful areas using a brainstem atlas (Coulombe et al., 2021), we appreciate that even with increased spatial resolution we are not able to precisely identify small brainstem nuclei. Thus, the described cluster locations should be evaluated in this light. To

extend these findings it would be of interest to explore whether a similar distribution of offset inhibitory and offset facilitatory responses arise with our shortened offset stimulus model in those with a chronic injury, as well as whether similar or disparate brainstem regions are responsible for the known impairment in offset analgesia in chronic pain states. Additionally, since we identified signal changes encompassing the NTS as related to offset inhibitory responses, future work should investigate whether autonomic markers such as heart rate variability or skin conductance thresholds are predictive or altered during the expression or offset analgesia. Such an association has been identified in placebo analgesia, where reduced low-frequency heart rate variability is associated with greater analgesic responses (Daniali and Flaten, 2020). Understanding the ensemble of biological and physiological changes that occur during endogenous pain modulation will assist in better contextualization of treatment responses. Moreover, since several of the brainstem areas we identify either receive dense input from; or densely innervate cortical areas to collectively enable the flexible modulation of pain, it would be of interest to employ voxel wise and effective connectivity analyses both between our identified brainstem regions and in cortical imaging space to determine the broader brain networks responsible for driving offset analgesia responses.

6. Conclusion

In this exploratory functional brain imaging investigation, using ultra-high field 7-Tesla fMRI, we demonstrate core nodes and activity changes in the brainstem descending modulatory system which are associated with offset analgesia and paradoxical hyperalgesia. Our significant group effect results indicate that offset analgesia may leverage a largely noradrenergic system, possibly being mechanistically distinct to other pain modulatory responses such as placebo analgesia, which are known to involve a canonical PAG-RVM pathway. Resolving the biological differences between differing pain modulatory mechanisms is critical for a wholistic understanding of the human capacity to influence one's own pain percept.

CRediT authorship contribution statement

Lewis S Crawford: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **James WM Kang:** Writing – review & editing, Visualization, Validation, Investigation. **Ashleigh H Wake:** Writing – review & editing, Validation, Data curation. **Thomas Love:** Writing – original draft, Data curation. **Paul M Macey:** Writing – review & editing, Software, Resources. **Kirsty Bannister:** Writing – review & editing, Resources, Methodology, Conceptualization. **Tor Wager:** Writing – review & editing, Resources, Conceptualization. **Vaughan G Macefield:** Writing – review & editing, Supervision, Project administration. **Kevin A Keay:** Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Luke A Henderson:** Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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National Health and Medical Research Council of Australia Grant 1130280. The authors declare no competing financial interests.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2026.121997.

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